



A crisis in cosmology? An overview of disagreements in the Hubble constant

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Over recent years, a disagreement has emerged between the ways of measuring the Hubble constant. This constant describes the expansion rate of the universe and is critical in estimating its age. To determine the Hubble constant we use the relation between a measured distance, along with a velocity from a measured redshift of the light. This literature review seeks to provide a brief history of the constant, an overview of processes for measuring cosmological distances to calculate the Hubble constant, and present proposed solutions to the emerging “Hubble tension.”

I. INTRODUCTION

Over the past 100 years, the Hubble constant H_0 has shifted quite significantly from its initial value. Today there are a handful of measured values, each with merit and uncertainty. Over recent years, however, those values began to no longer agree with each other, leading to the “Hubble tension,” or a “crisis in cosmology.” The goal of this review is to provide an overview of the past and present of the constant, and how the varying ways of measuring cosmic distances contribute to the current debate.

II. EDWIN HUBBLE’S INFLUENCE ON H_0

The Hubble constant, denoted as H_0 represents the rate at which the universe is expanding. The exploration of its value has a rich history, marked by pioneering observations and significant developments in cosmology.

In the 1920s, using the Hooker Telescope at Mount Wilson Observatory, Edwin Hubble observed distant galaxies (known at the time as “spiral nebulae”). Due to the resolving power of the new telescope’s 100-inch lens, Hubble was able to identify Cepheid variable stars in the objects, determining that they were outside of the Milky Way [1]. Not long after determining this, Hubble proposed classifications for the objects based on their shape [2].

Later, he noticed a correlation between their redshifts (and identifier of velocity) and distances. This led to the formulation of Hubble’s Law (later known as the Hubble–Lemaître Law), expressing the linear relationship between a galaxy’s velocity v (due to expansion) and its distance d :

$$v = H_0 d.$$

Hubble’s early estimate for the Hubble constant was around 500 km/s/Mpc [3], but this was based on less ac-

curate distance measurements and had large uncertainties. The foundational concept, however, paved the way for future investigations.

Years after Hubble’s contributions, the Hubble Space Telescope (HST), launched in 1990, emerged as an instrumental platform for refining measurements of H_0 . Enhanced precision in distance determinations, facilitated by the HST, contributed significantly to elucidating the Hubble constant [4].

III. COSMIC DISTANCE LADDER CONCEPTS

The cosmic distance ladder serves as a hierarchical approach to ascertain distances to celestial objects, ranging from nearby astronomical bodies to those situated at cosmological distances. This methodology integrates various measurement techniques, each calibrated against the preceding tier, ensuring a coherent and accurate estimation of distances in the vastness of the cosmos.

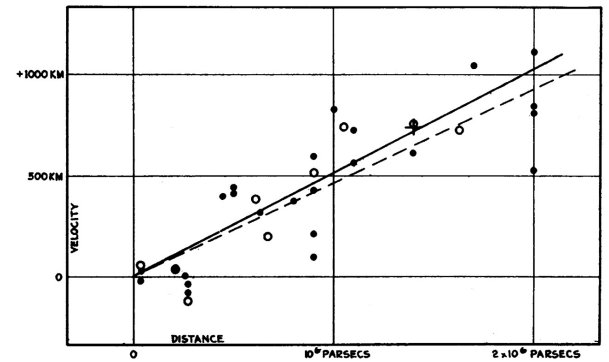


FIG. 1. Hubble’s original plot, the first of the now famous “Hubble diagrams.” Distance in parsecs is on the horizontal axis, while velocity derived from redshift is on the vertical axis.

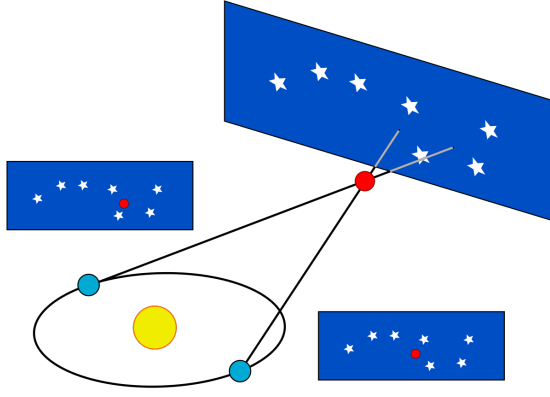


FIG. 2. Stellar parallax motion from annual parallax. Half the apex angle is the parallax angle.
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A. Parallax of Nearby Stars

The foundational rung of the cosmic distance ladder involves the measurement of stellar parallax. As the Earth orbits the Sun, nearby stars exhibit apparent shifts in position against the background fig. 1. By measuring this change in angle p (in arcseconds), astronomers can determine distances to stars within a few hundred parsecs (3.26 ly):

$$d = \frac{1}{p}$$

The maximum distances that can reasonably be measured using parallax methods have been extended over the years by many different efforts [5–7]. For some additional reading on the mathematics of parallax, I recommend reading *Mathematics of Statistical Parallax and the Local Distance Scale* [8].

B. Main Sequence Fitting

Moving beyond parallax, the cosmic distance ladder extends to calibrating distances to star clusters through main sequence fitting. This method leverages the relationship between a star's luminosity and its color, allowing astronomers to estimate the distance to a star cluster based on its observed characteristics. Often used to help visualize this color-luminosity relation is the famous Hertzsprung-Russell diagram fig. 3.

C. Eclipsing Binaries

Eclipsing binary star systems present another avenue for refining distance measurements and contributing to

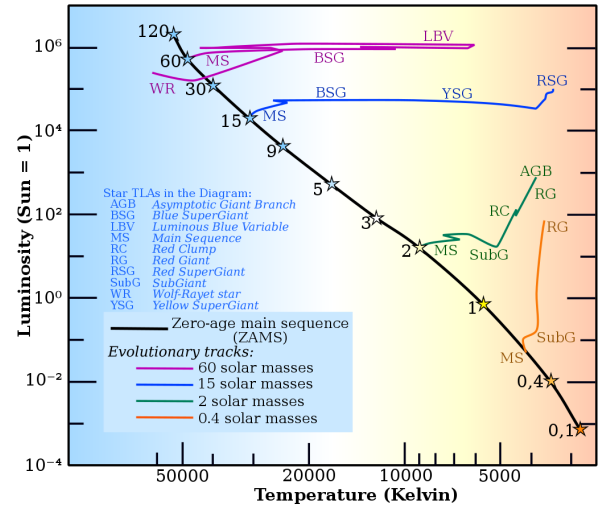


FIG. 3. This diagram shows both the color-luminosity relations of main sequence stars and the evolution of stars from the main sequence.
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the determination of the Hubble constant. Eclipsing binaries are binary star systems aligned in such a way that one star periodically passes in front of the other, causing observable variations in brightness.

The periodic eclipses result in characteristic light curves—plots of brightness versus time. By analyzing these light curves, astronomers can deduce critical information about the system, including the size, mass, and orbital parameters of the stars. Eclipsing binaries serve as geometric distance indicators, providing a direct measure of the distance to the system.

Eclipsing binaries play a vital role in calibrating the cosmic distance ladder [9]. Their proximity allows for accurate distance measurements within our Milky Way galaxy. The knowledge gained from these nearby systems serves as a crucial anchor point for calibrating other distance indicators, contributing to the precision of the entire ladder.

D. Cepheid Variable Stars

Cepheid variable stars play a pivotal role in the cosmic distance ladder and are instrumental in refining our understanding of the Hubble constant. These stars undergo periodic changes in brightness, and their pulsation period is directly related to their intrinsic luminosity.

This intrinsic brightness-luminosity relationship forms the basis for using Cepheids as standard candles for distance measurements. By observing the period P (in days) of a Cepheid variable, astronomers can deduce its intrinsic brightness and, consequently, using the formula for absolute magnitude [10],

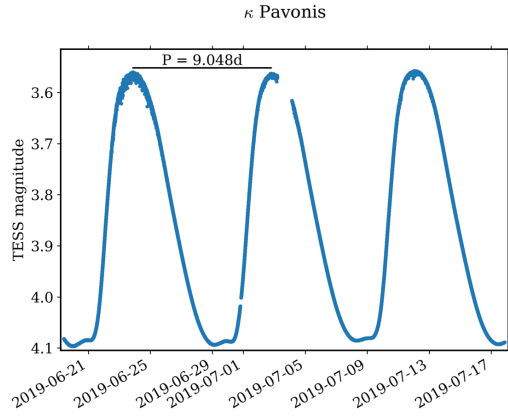


FIG. 4. Lightcurve of the W Virginis (Type II Cepheid) variable κ Pavonis recorded by NASA's Transiting Exoplanet Survey Satellite (TESS).

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$$\langle M_V \rangle = -2.77 \log(P) - 1.44 \pm 0.05,$$

determine its distance using the magnitude-distance relation:

$$d = 10^{\frac{5 + m - M_V}{5}}.$$

E. Tip of the Red Giant Branch (TRGB)

The Tip of the Red Giant Branch (TRGB) serves as a crucial astronomical yardstick for measuring cosmic distances, particularly to nearby galaxies. This method capitalizes on the distinctive feature seen in the color-magnitude diagram of stellar populations, providing a reliable and consistent reference point for distance determination.

The TRGB represents the luminosity peak of stars evolving off the main sequence and ascending the red giant branch. This specific point in the color-magnitude diagram corresponds to the transition from hydrogen burning to helium shell burning in these aging stars. The luminosity of the TRGB is relatively constant, making it a valuable standard candle for distance measurements.

Observationally, astronomers target the color-magnitude diagram of a stellar population, identifying the point where the luminosity function exhibits a sharp change—indicating the TRGB. The absolute magnitude of the TRGB can be calibrated through parallax measurements of nearby stars with known distances, providing a standardized reference for more distant systems.

The TRGB method offers several advantages in the context of distance measurements. Unlike some other

standard candles, TRGB is present in a variety of stellar populations, allowing for its application to a wide range of galaxies. It is particularly well-suited for the study of resolved stellar populations in nearby galaxies, contributing to the calibration of the cosmic distance ladder.

In the quest to determine the Hubble constant, TRGB plays a vital role in calibrating distances to galaxies within the Local Group and beyond. Its application extends the reach of distance measurements, bridging the gap between nearby systems and more distant extragalactic objects.

F. Type Ia Supernovae

One of the distinctive features of Type Ia supernovae is their remarkably uniform luminosity. The explosion releases a consistent amount of energy, making SNe Ia invaluable for distance measurements. The predictable brightness allows astronomers to use these events as standard candles [11], assuming that their intrinsic luminosity is relatively constant.

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The light curve, a plot of brightness versus time, of a Type Ia supernova exhibits a characteristic shape fig. 5. The rise and fall of luminosity over time follow a well-defined pattern, with a consistent peak brightness. Analyzing the light curve provides astronomers with crucial information about the supernova's intrinsic properties, allowing for accurate distance determinations.

Calibrating the luminosity of Type Ia supernovae involves comparing their observed brightness to their intrinsic brightness. This calibration is often carried out

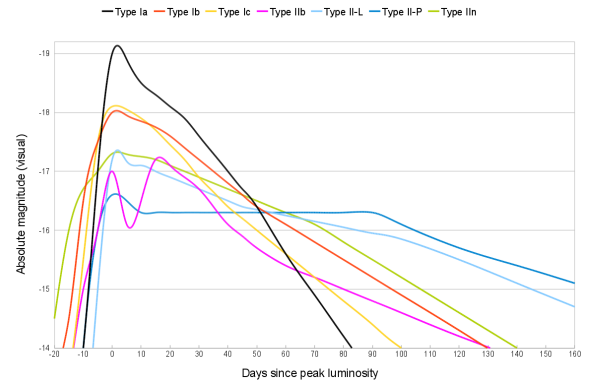


FIG. 5. This image shows light curves of the main supernova types, representative of peak brightness, rise times, and decay times.

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using nearby supernovae with well-constrained distances, allowing for the establishment of a standardized relationship between luminosity and distance.

Type Ia supernovae are particularly potent tools for measuring cosmic distances to galaxies at cosmological scales. Their uniform luminosity makes them visible across vast distances, enabling astronomers to probe the expanses of the universe. Measurements of the apparent brightness of SNe Ia, when compared to their calibrated intrinsic brightness, contribute crucial data points to the cosmic distance ladder and the determination of the Hubble constant.

Type Ia supernovae played a pivotal role in the discovery of the accelerated expansion of the universe. Observations of distant supernovae by research teams such as the Supernova Cosmology Project and the High-Z Supernova Search Team provided key evidence for the existence of dark energy, earning the 2011 Nobel Prize in Physics.

While Type Ia supernovae offer a powerful tool for cosmological studies, systematic uncertainties exist. Variations in the progenitor systems, such as differences in metallicity or the nature of the companion star, can introduce subtle variations in the brightness of these events. Addressing and minimizing these uncertainties is an ongoing focus of research.

G. Gravitational Waves

In recent years, the detection of gravitational waves has emerged as a promising avenue for measuring the Hubble constant. Gravitational waves, ripples in spacetime caused by the acceleration of massive objects, provide a unique and direct means to probe cosmic distances.

Binary neutron star and neutron star-black hole mergers serve as “standard sirens” in the context of gravitational wave astronomy [12, 13]. These events emit both gravitational waves and electromagnetic signals, allowing for a simultaneous measurement of the distance to the source.

By combining the observed gravitational wave signal with electromagnetic data, astronomers can determine the luminosity distance to the merging system. This provides a direct and independent measure of the distance, making gravitational waves a unique tool for calibrating cosmic distances.

H. Redshift and the Hubble–Lemaître Law

At larger cosmological scales, the expansion of the universe becomes a guiding principle. The redshift of galaxies, observed through spectroscopy, provides a measure of their recession velocity. The Hubble–Lemaître Law relates this velocity to distance, offering a means to estimate distances to galaxies across the cosmos.

I. Cosmic Microwave Background

The Cosmic Microwave Background (CMB) is a remnant radiation that permeates the entire universe, originating from the hot, dense state of the early universe. It provides a unique and powerful tool for understanding the large-scale structure, composition, and expansion of the cosmos.

One of the defining characteristics of the CMB is its remarkable isotropy and homogeneity. The radiation is nearly uniform in all directions, with tiny fluctuations representing the seeds of cosmic structure. These fluctuations encode valuable information about the conditions of the early universe and serve as a fundamental reference frame for cosmological studies.

The CMB carries imprints of acoustic oscillations that occurred in the primordial plasma before the universe became transparent. These oscillations create distinct patterns in the temperature and polarization of the CMB, commonly observed in experiments like the Planck satellite [14]. Analyzing these patterns allows scientists to extract precise information about the density, geometry, and composition of the universe.

The CMB is a key player in determining the Hubble constant, especially through measurements of the sound horizon at the time of photon decoupling. The acoustic oscillations in the early universe create characteristic features in the CMB power spectrum, and the angular scale of these features is intimately connected to the expansion rate of the universe at that time.

IV. CURRENT DISAGREEMENTS

In early measurements of the Hubble constant, the uncertainties in the values included each other. However, over recent years, there has emerged a disagreement in local and distant measurements fig. 6.

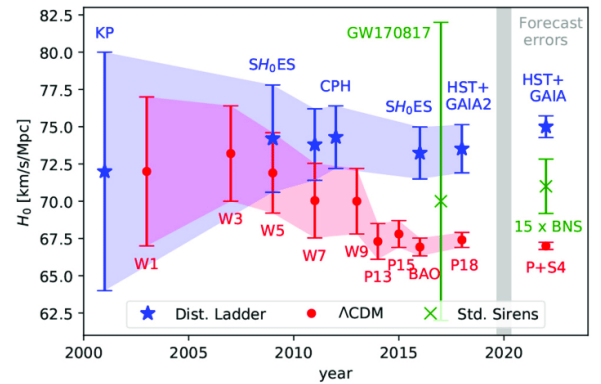


FIG. 6. This diagram shows the growing disparity between measurement techniques, including an initial value determined using standard sirens.

Source: [15]

A. Local Measurements vs. Cosmic Microwave Background (CMB)

The prominent source of tension arises from disparities between local measurements of H_0 and those derived from observations of the cosmic microwave background. Local measurements, employing techniques like Cepheid variables, TRGB, and Type Ia supernovae, yield values around 73-74 km/s/Mpc. However, analyses of the CMB, often through studies like the Planck satellite mission, suggest a lower value, typically around 67 km/s/Mpc. This discrepancy has led to what is colloquially known as the “Hubble tension.”

B. A Closed Universe

There are three main shapes for spacetime to exist as. They can be described by how two parallel lines would behave in those environments:

- **Open Universe:** parallel lines diverge from each other.
- **Flat Universe:** parallel lines remain parallel and never meet.
- **Closed Universe:** parallel lines converge towards each other.

A recent study has shown evidence that the universe is slightly closed, rather than flat or open [16]. The authors note that this possibly adds to the current tension in H_0 , though the results are preliminary.

V. PROPOSED SOLUTIONS

Though studies a few years ago claimed that the tension would be resolved with better data or larger telescopes [17], recent efforts have shown that may not be enough to resolve it.

A. Correcting the Cepheid Measurements

Within the realm of local measurements, the calibration of Cepheid variables plays a crucial role. Different studies employing Cepheids in the nearby universe may yield slightly different results. The calibration of these standard candles introduces uncertainties that contribute to the overall tension in H_0 measurements. Over time, the idea evolved that this may be due to the stars being “biased bright” by nearby stars. After the launch of the James Webb Space Telescope, many hoped that the ability to better resolve individual stars would mitigate this. However, many different studies from varying angles have now shown this bias to not contribute to the tension.

Preliminary work on correcting the Cepheids using Hubble data showed only a 0.23% bias bright [18]. A similar conclusion followed for JWST data [19, 20]. Later, more robust studies using Hubble additionally confirmed such, continuing the tension [21], with some follow-up studies even increasing the tension [22]. A very recent study using JWST data has shown an 8σ confidence that Cepheids are not biased bright, a “greater confidence than that of the Hubble Tension itself” [23].

B. Utilizing TRGB Measurements

Another proposed solution was to use TRGB measurements in place of or to correct the Cepheids measurements [24]. The reasoning behind this is the history and physical parallels between the two measurement systems [25, 26]. However, as with the previously proposed solution, later studies using TRGB measurements also increased the tension, as they were shown to agree with the Cepheid measurements [27, 28].

C. Superclusters and Voids

The Laniakea supercluster, discovered in 2014 [29], is part of another proposed solution to the tension. The idea is that the motion of local galaxies within the cluster may be skewing the measurements. A recent study analyzed this idea and found it to also worsen the tension [30]. The authors do admit that the model is likely too oversimplified to contribute any meaningful conclusions, and they are working towards more accurate ways of modeling the cluster. In addition to this idea, cosmic voids have also been proposed to cause a similar skewing of measurements through the same type of gravitational interactions.

VI. CONCLUSION

The Hubble constant has had a long history and this most recent decade has been as tumultuous as its first. As measured quantities’ uncertainties began to shrink, they no longer overlapped, and the “Hubble tension” began to arise. Though many hoped for the Hubble tension to be resolved over the years through various means, whether that be better equipment or more precise data, those same solutions instead worsened the so-called “crisis in cosmology”. Cepheid variable and TRGB measurements became more confidently disconnected from CMB measurements, with seemingly each successive study widening the gap. One recent study has shown an 8σ confidence that Cepheids are not biased bright, a “greater confidence than that of the Hubble Tension itself” [23]. One exciting possibility given this trend is that there are new models and physics to discover. What happens in the next 5, 10, 100 years, whatever it may be, will likely

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